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SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME:Computer Networks COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

79%

Case study

QUESTIONS:4 Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations.**
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution	20	Innovative, feasible solutions with	Logical solutions with	Basic solutions	Incorrect or no

MARK ALLOCATION

	Understanding of Case (25%)	Application of Concepts(25%)	Analysis(20%)	Solution Design(20%)	Clarity (10%)
Q1 8	2.5	2.5	1	1	1
Q2 9	2.5	2.5	2	1	1
Q3 8.5	2	2	2	2	0.5
Q4 6	1.5	1	1	2	0.5
~~Q5~~					

$\frac{31.5}{40}$

SIMATS ENGINEERING

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2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

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Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

MARK ALLOCATION

	Understanding of Case **(25%)**	**Application of Concepts (25%)**	**Analysis (20%)**	**Solution Design (20%)**	**Clarity (10%)**
Q1					
Q2					
Q3					
Q4					
Q5					

Case Study 1: 1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

Cloud-Based Video Conferencing:
The company is experiencing **8% packet loss, 180 ms jitter, fluctuating bandwidth utilization, voice distortion, frozen video frames, and delayed communication** during peak hours. The case identifies the transport protocol and application-level buffering strategy as possible causes.

1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The problem is primarily related to the **transport protocol characteristics and application-level handling of real-time traffic**.
For video conferencing, **UDP is generally more suitable than TCP** because UDP does not wait for retransmission of lost packets. With 8% packet loss and 180 ms jitter, using TCP can introduce additional delay because TCP retransmits lost packets and performs congestion control. This can result in delayed audio and video.

However, the application layer also has an important role. If the application's buffering mechanism is too large, it can increase delay. If the buffer is too small, packet loss and jitter can cause frozen video and distorted audio.
Therefore, the issue is a **combination of network/transport conditions and application-level buffering**, with real-time transport behavior being the major concern.

2. Justify your answer based on the communication requirements of real-time multimedia applications.

Real-time multimedia applications require:

- **Low latency** for immediate communication.
- **Low jitter** to maintain smooth audio and video.
- **Low packet loss** to prevent distortion and missing frames.
- Continuous data delivery rather than perfect retransmission.

- Efficient use of available bandwidth.

TCP prioritizes reliable and ordered delivery. This is useful for applications such as web pages and file transfers, but retransmissions can increase delay.
UDP does not guarantee delivery or ordering, but it avoids retransmission delays. Therefore, UDP is generally better suited for interactive voice and video where **timeliness is more important than perfect delivery**.

3. Recommend suitable protocol-level or application-level improvements.
The following improvements can enhance conferencing quality:

1. Use **UDP-based real-time communication** where appropriate.
2. Apply **adaptive bitrate control** so video quality automatically adjusts according to available bandwidth.
3. Use **jitter buffers** to smooth variations in packet arrival time.
4. Keep buffering small enough to avoid excessive latency.
5. Apply **packet-loss concealment** for missing audio/video packets.
6. Prioritize voice and video traffic using **QoS mechanisms**.
7. Monitor and control congestion during peak business hours.
8. Use suitable error-resilience techniques for real-time media.

Case Study 2: A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

Global Web Services and DNS:
The company operates web services across multiple continents and experiences an average **DNS lookup time of approximately 800 ms**. It plans to redesign DNS infrastructure using Anycast DNS, DNS pre-fetching, and TTL optimization while maintaining continuous availability.

1. Analyze the reasons behind the high DNS resolution time.
The 800 ms DNS resolution time may be caused by several factors:

- DNS servers may be geographically far from users.
- Multiple DNS resolution steps may be required.
- DNS records may not be available in local caches.
- Low TTL values may cause frequent DNS queries.
- DNS servers may be overloaded.

- Network congestion can increase query-response time.
- A centralized DNS infrastructure can create additional latency and a potential point of failure.

Therefore, inefficient DNS server placement, insufficient caching, and repeated DNS resolution can significantly increase page-loading time.

2. Propose a suitable DNS architecture to reduce lookup time below 50 ms.
A suitable architecture is a **distributed Anycast DNS architecture**.
Architecture:

User → Nearest Anycast DNS Server → Cached DNS Response → Application Server
Multiple DNS servers should be deployed across different geographical regions. All servers can advertise the same Anycast IP address. Routing then directs users toward a nearby DNS server.

The architecture should include:
Anycast DNS:
- Deploy DNS servers at multiple locations.
- Advertise the same IP address from each location.
- Route users to the nearest available DNS node.

DNS Caching:
- Cache frequently requested DNS records at recursive resolvers.
- Reduce repeated queries to authoritative servers.

DNS Pre-fetching:
- Frequently accessed DNS records can be resolved before they are required.
- This reduces waiting time for subsequent requests.

TTL Optimization:
- Use an appropriate TTL so records remain cached for a reasonable period.
- Avoid excessively low TTL values that cause unnecessary DNS queries.
- Avoid excessively high TTL values when rapid changes are required.

With geographically distributed servers, caching, pre-fetching, and optimized TTL values, DNS resolution latency can be significantly reduced, with the target being **below 50 ms**.

3. Evaluate the advantages and limitations.

Aspect	**Advantages**	**Limitations**
Performance	Reduces DNS lookup latency	Requires multiple DNS locations
Scalability	Handles traffic from different regions	Infrastructure cost increases
Availability	Failure of one DNS node can be handled by another	Requires reliable routing
Anycast	Directs users toward nearby DNS service	Routing problems can affect availability

Aspect	Advantages	Limitations
Caching	Reduces repeated DNS queries	Cached data can become outdated
Pre-fetching	Reduces lookup delay	Generates additional DNS activity
TTL optimization	Balances freshness and caching	Incorrect TTL values can cause problems

Overall, the proposed architecture provides better **performance, scalability, and fault tolerance** than a centralized DNS system.

Case Study 3: A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

1. Examine the impact of centralized DNS architecture on application performance.
2. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

Centralized DNS and Global Cloud Services:
The cloud provider serves millions of users worldwide. During promotional events, DNS queries take nearly **800 ms because all requests are directed to a single primary DNS server**. The company requires a highly available global DNS solution.

4. Examine the impact of centralized DNS architecture on application performance.
A centralized DNS architecture creates a **single point of dependency** for DNS resolution.
When millions of users send requests to one DNS server:

- The server can become overloaded.
- DNS query queues can increase.
- Response time increases.
- Application startup/loading is delayed.
- Network traffic becomes concentrated at one location.
- Server failure can make services difficult or impossible to access.
- Promotional events can cause sudden traffic spikes and DNS congestion.

The reported **800 ms DNS lookup delay** therefore directly affects application performance because users must complete DNS resolution before establishing communication with the required cloud service.

5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

A suitable design is:

Users Worldwide
↓
Anycast DNS IP
↓
Nearest DNS Server
↓
Local/Recursive Cache
↓
Authoritative DNS Servers
↓
Cloud Application

The company should:

1. Deploy multiple DNS servers across major geographical regions.
2. Use **Anycast routing** to direct users to nearby DNS servers.
3. Maintain redundant DNS servers in each region.
4. Use DNS caching to reduce repeated queries.
5. Use suitable TTL values for frequently accessed records.
6. Monitor DNS servers continuously.
7. Automatically route traffic away from failed DNS nodes.
8. Distribute DNS traffic during promotional events.

This removes the single-server bottleneck and improves availability.

6. Analyze how TTL values, Anycast routing, and DNS caching influence performance and fault tolerance.

TTL:

TTL determines how long DNS information can remain cached. A suitable TTL reduces the number of DNS queries and improves performance. However, very high TTL values can delay propagation of DNS changes.

Anycast Routing:

Anycast allows multiple DNS servers to use the same IP address. Users are generally routed toward a nearby available DNS location. This reduces latency and provides redundancy.

DNS Caching:

Caching stores DNS responses so repeated requests can be answered without contacting the authoritative DNS server every time. This reduces DNS traffic and lookup time.

Together, these mechanisms improve **performance, scalability, and fault tolerance**.

Case Study 4: A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

TCP Latency in Electronic Stock Trading:

The financial institution experiences an increase in order acknowledgment latency from **1 ms to 40 ms**. Engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering. The case specifically identifies **flow control, Nagle's algorithm, and SYN queue limitations** for analysis.

1. Analyze three possible TCP-related factors responsible for the latency increase.

a) Flow Control:

TCP uses a receive window to prevent a sender from transmitting more data than the receiver can handle.

If the receiver's buffer becomes full or the advertised window becomes small, the sender may have to slow down or wait.

In a high-frequency trading environment, this can increase the time required to process and acknowledge transactions.

b) Nagle's Algorithm:

Nagle's algorithm combines small TCP segments into larger segments to reduce network overhead.
Although this improves efficiency for some applications, it can introduce additional delay when applications send many small messages.
In stock trading, where small transactions require very fast responses, this delay can be undesirable.

c) SYN Queue Limitations:
TCP requires a connection establishment process using the **SYN, SYN-ACK, and ACK** exchange.
During market opening, a very large number of connections may arrive simultaneously. If the server's SYN queue becomes overloaded, connection establishment can be delayed or connections may be dropped.
This increases the time before trading requests can be processed.

2. Explain how each factor affects transaction processing in high-frequency systems.

TCP Factor	Effect on Trading System
Flow control	Limits transmission when receiver buffers are congested
Nagle's algorithm	May delay small transaction messages
SYN queue limitations	Delays new TCP connection establishment
Retransmissions	Increase latency when packets are lost
Excessive buffering	Creates additional waiting time

In high-frequency systems, even a few milliseconds can affect transaction processing. Therefore, TCP configuration and network resources must be optimized for low latency.

3. Recommend appropriate TCP configuration changes:
The following changes can help reduce latency:

1. **Disable Nagle's algorithm** using TCP_NODELAY for latency-sensitive small messages where appropriate.
2. Increase and properly configure **TCP receive and send buffers** to avoid unnecessary flow-control stalls.
3. Increase the server's **SYN backlog/queue capacity** to handle bursts of connection requests.
4. Use connection reuse or persistent connections to reduce repeated TCP handshakes.
5. Investigate packet loss and retransmissions and improve network reliability.
6. Monitor TCP congestion and buffer utilization.
7. Use suitable QoS policies to prioritize latency-sensitive trading traffic.
8. Optimize server and network resources to handle market-opening traffic spikes.

These changes can reduce unnecessary delays while maintaining TCP reliability.